

Bias–Variance Decomposition for Regularised Laplace Inversion Surrogates

Abstract

We derive rigorously the bias–variance decomposition of the total reconstruction error for a Laplace-domain neural surrogate model combined with a regularised inversion step. The main result is a bound of the form

$$\mathcal{E}_{\text{tot}} \leq \frac{\mathbb{E}_{\theta}[\|(\alpha_t \mathbf{D}_t^* \mathbf{D}_t + \lambda \mathbf{I}) \mathbf{v}\|]}{\sigma_{\min}(\mathbf{A})} + \frac{\|\mathbf{F}\|}{\sigma_{\min}(\mathbf{A})} \cdot \mathbb{E}_{\theta}[\|\hat{\mathbf{u}} - \hat{\mathbf{u}}^{\text{pred}}\|],$$

where $\sigma_{\min}(\mathbf{A})$ denotes the smallest singular value (or equivalently, the smallest eigenvalue) of the normal equation matrix $\mathbf{A}$. The first term is a *regularisation bias* (representing the intrinsic transformation and regularisation error) and the second term is the *variance* (the surrogate error propagated through the inversion operator). Both terms are rigorous consequences of the linear structure of the problem.

Contents

1	Discrete Setting	2
1.1	Notation	2
1.2	Laplace transform matrix	2
1.3	Regularised inversion	2
2	Normal Equations and Closed-Form Solution	3
2.1	The operator $\mathbf{A}$	3
3	Bias–Variance Decomposition	3
3.1	Splitting the error	3
3.2	Simplifying the bias term	3
3.3	Bounding each term	4
3.4	Main result	4
4	Interpretation	4
4.1	Role of each term	4
4.2	What remains to be bounded	5
5	Comparison with the Bromwich–FFT Pipeline	5
5.1	Bromwich contour and FFT structure	5
5.2	Truncation to $k_{\max}$ frequencies	6
5.3	The FFT inversion is not the regularised inverse	7
5.4	Summary comparison	7

1 Discrete Setting

1.1 Notation

Let $\Omega \subset \mathbb{R}^d$ be the spatial domain, discretised on N_x nodes $\{x_j\}_{j=1}^{N_x}$. Time is discretised on a uniform grid $t_n = n\Delta t$, $n = 0, \dots, N_t - 1$, with time step $\Delta t > 0$ and final time $T = (N_t - 1)\Delta t$.

For each simulation parameter $\theta \in \Theta$, the spatio-temporal field is represented as a matrix

$$\mathbf{V}(\theta) \in \mathbb{R}^{N_x \times N_t}, \quad V_{j,n}(\theta) = U(x_j, t_n; \theta).$$

Because the regularised inversion decouples over spatial nodes (when only temporal regularisation is used), we work with a single spatial node and drop the index j throughout. The field at one node is a vector $\mathbf{v}(\theta) \in \mathbb{R}^{N_t}$.

1.2 Laplace transform matrix

Assumption 1.1 (Coercivity). *The underlying spatial operator is strongly elliptic or coercive, meaning there exists a constant $\lambda_1 > 0$ such that the system's eigenvalues have real parts bounded above by $-\lambda_1$. This ensures the field decays exponentially and its Laplace transform is well-defined for $\text{Re}(s) > -\lambda_1$.*

Fix N complex frequencies $s_1, \dots, s_N \in \mathbb{C}$ with $\text{Re}(s_k) > -\lambda_1$ for all k , where $\lambda_1 > 0$ is the coercivity constant of the spatial operator (see Assumption 1.1).

Definition 1.1 (Laplace matrix). *The Laplace matrix $\mathbf{F} \in \mathbb{C}^{N \times N_t}$ is defined by*

$$F_{k,n} = \Delta t \cdot w_n \cdot e^{-s_k t_n}, \quad k = 1, \dots, N, \quad n = 0, \dots, N_t - 1, \quad (1)$$

where w_n are quadrature weights (trapezoidal: $w_0 = w_{N_t-1} = \frac{1}{2}$, $w_n = 1$ otherwise).

The discrete Laplace transform of $\mathbf{v}$ is then

$$\hat{\mathbf{u}} = \mathbf{F}\mathbf{v} \in \mathbb{C}^N. \quad (2)$$

Each component $\hat{u}_k = (\mathbf{F}\mathbf{v})_k$ approximates $\hat{U}(s_k; \theta) = \int_0^T U(t; \theta) e^{-s_k t} dt$.

1.3 Regularised inversion

The surrogate predicts $\hat{\mathbf{u}}^{\text{pred}} \in \mathbb{C}^N$. Given these predictions, we reconstruct $\mathbf{v}^*$ by solving the regularised least-squares problem

$$\mathbf{v}^*(\theta) = \arg \min_{\mathbf{v} \in \mathbb{R}^{N_t}} \left\| \mathbf{F}\mathbf{v} - \hat{\mathbf{u}}^{\text{pred}}(\theta) \right\|_{\mathbb{C}^N}^2 + \alpha_t \|\mathbf{D}_t \mathbf{v}\|^2 + \lambda \|\mathbf{v}\|^2, \quad (3)$$

where:

- $\alpha_t \geq 0$ is a temporal smoothing parameter,
- $\lambda > 0$ is a Tikhonov regularisation parameter,
- $\mathbf{D}_t \in \mathbb{R}^{(N_t-1) \times N_t}$ is the first-difference operator:

$$(\mathbf{D}_t \mathbf{v})_n = v_{n+1} - v_n, \quad n = 0, \dots, N_t - 2.$$

Remark 1.2. *The constraint $\mathbf{v} \in \mathbb{R}^{N_t}$ (real-valued reconstruction) is handled by splitting $\mathbf{F} = \text{Re}(\mathbf{F}) + i \text{Im}(\mathbf{F})$ and forming the real augmented system*

$$\begin{pmatrix} \text{Re}(\mathbf{F}) \\ \text{Im}(\mathbf{F}) \end{pmatrix} \mathbf{v} \approx \begin{pmatrix} \text{Re}(\hat{\mathbf{u}}^{\text{pred}}) \\ \text{Im}(\hat{\mathbf{u}}^{\text{pred}}) \end{pmatrix}.$$

For clarity of exposition we write the analysis in complex form; the real-valued case follows identically by replacing $\mathbb{C}^N$ with $\mathbb{R}^{2N}$.

2 Normal Equations and Closed-Form Solution

2.1 The operator $\mathbf{A}$

The optimality condition (normal equations) for (3) is:

$$\begin{aligned} 0 &= \frac{\partial}{\partial \mathbf{v}} \left[\left\| \mathbf{F}\mathbf{v} - \hat{\mathbf{u}}^{\text{pred}} \right\|^2 + \alpha_t \|\mathbf{D}_t \mathbf{v}\|^2 + \lambda \|\mathbf{v}\|^2 \right] \\ &= 2\mathbf{F}^*(\mathbf{F}\mathbf{v} - \hat{\mathbf{u}}^{\text{pred}}) + 2\alpha_t \mathbf{D}_t^* \mathbf{D}_t \mathbf{v} + 2\lambda \mathbf{v}. \end{aligned}$$

Rearranging:

$$\underbrace{(\mathbf{F}^* \mathbf{F} + \alpha_t \mathbf{D}_t^* \mathbf{D}_t + \lambda \mathbf{I})}_{=: \mathbf{A}} \mathbf{v}^* = \mathbf{F}^* \hat{\mathbf{u}}^{\text{pred}}. \quad (4)$$

Proposition 2.1 (Invertibility of $\mathbf{A}$). *If $\lambda > 0$, then $\mathbf{A}$ is symmetric positive definite, hence invertible.*

Proof. Each of $\mathbf{F}^* \mathbf{F}$, $\mathbf{D}_t^* \mathbf{D}_t$, and $\lambda \mathbf{I}$ is positive semi-definite. Since $\lambda > 0$, for any $\mathbf{v} \neq \mathbf{0}$:

$$\mathbf{v}^\top \mathbf{A} \mathbf{v} \geq \lambda \|\mathbf{v}\|^2 > 0.$$

Hence $\mathbf{A} \succ 0$ and its smallest singular value (which coincides with its smallest eigenvalue) satisfies $\sigma_{\min}(\mathbf{A}) \geq \lambda > 0$. $\square$

Remark 2.2. When $\lambda = 0$ and $\alpha_t > 0$, invertibility still holds provided $s_k \neq 0$ for all k . Indeed, $\ker(\mathbf{A}) \subseteq \ker(\mathbf{F}) \cap \ker(\mathbf{D}_t)$; the latter forces $\mathbf{v} = c\mathbf{1}$ (constant), and $\mathbf{F}(c\mathbf{1}) = 0$ would require $\sum_n w_n e^{-s_k t_n} = 0$ for all k , which fails when $s_k \neq 0$. We keep $\lambda > 0$ for simplicity.

The unique solution of (4) is therefore:

$$\boxed{\mathbf{v}^*(\theta) = \mathbf{A}^{-1} \mathbf{F}^* \hat{\mathbf{u}}^{\text{pred}}(\theta)}. \quad (5)$$

3 Bias–Variance Decomposition

3.1 Splitting the error

We decompose the total error by adding and subtracting the intermediate quantity $\mathbf{A}^{-1} \mathbf{F}^* \hat{\mathbf{u}}(\theta)$, where $\hat{\mathbf{u}}(\theta) = \mathbf{F}\mathbf{v}(\theta)$ are the *exact* Laplace coefficients:

$$\begin{aligned} \mathbf{v}(\theta) - \mathbf{v}^*(\theta) &= \mathbf{v}(\theta) - \mathbf{A}^{-1} \mathbf{F}^* \hat{\mathbf{u}}^{\text{pred}}(\theta) \\ &= \underbrace{\mathbf{v}(\theta) - \mathbf{A}^{-1} \mathbf{F}^* \mathbf{F} \mathbf{v}(\theta)}_{\substack{\text{bias term} \\ \text{(transformation error)}}} + \underbrace{\mathbf{A}^{-1} \mathbf{F}^* (\hat{\mathbf{u}}(\theta) - \hat{\mathbf{u}}^{\text{pred}}(\theta))}_{\substack{\text{variance term} \\ \text{(surrogate error)}}}. \end{aligned} \quad (6)$$

3.2 Simplifying the bias term

Using the definition $\mathbf{A} = \mathbf{F}^* \mathbf{F} + \alpha_t \mathbf{D}_t^* \mathbf{D}_t + \lambda \mathbf{I}$:

$$\begin{aligned} \mathbf{v} - \mathbf{A}^{-1} \mathbf{F}^* \mathbf{F} \mathbf{v} &= \mathbf{A}^{-1} \mathbf{A} \mathbf{v} - \mathbf{A}^{-1} \mathbf{F}^* \mathbf{F} \mathbf{v} \\ &= \mathbf{A}^{-1} (\mathbf{A} - \mathbf{F}^* \mathbf{F}) \mathbf{v} \\ &= \mathbf{A}^{-1} (\alpha_t \mathbf{D}_t^* \mathbf{D}_t + \lambda \mathbf{I}) \mathbf{v}. \end{aligned} \quad (7)$$

The bias depends only on the *true* field $\mathbf{v}$ and the regularisation operators — it is independent of the surrogate predictions.

3.3 Bounding each term

Lemma 3.1 (Bound on the bias).

$$\|\mathbf{v} - \mathbf{A}^{-1}\mathbf{F}^*\mathbf{F}\mathbf{v}\| \leq \frac{\|(\alpha_t\mathbf{D}_t^*\mathbf{D}_t + \lambda\mathbf{I})\mathbf{v}\|}{\sigma_{\min}(\mathbf{A})}. \quad (8)$$

Proof. From (7) and the sub-multiplicativity of the operator norm:

$$\|\mathbf{A}^{-1}(\alpha_t\mathbf{D}_t^*\mathbf{D}_t + \lambda\mathbf{I})\mathbf{v}\| \leq \|\mathbf{A}^{-1}\| \cdot \|(\alpha_t\mathbf{D}_t^*\mathbf{D}_t + \lambda\mathbf{I})\mathbf{v}\|.$$

Since $\mathbf{A} \succ 0$, $\|\mathbf{A}^{-1}\| = 1/\sigma_{\min}(\mathbf{A})$, which gives (8). $\square$

Lemma 3.2 (Bound on the variance).

$$\|\mathbf{A}^{-1}\mathbf{F}^*(\hat{\mathbf{u}} - \hat{\mathbf{u}}^{\text{pred}})\| \leq \frac{\|\mathbf{F}\|}{\sigma_{\min}(\mathbf{A})} \cdot \|\hat{\mathbf{u}} - \hat{\mathbf{u}}^{\text{pred}}\|_{\mathbb{C}^N}. \quad (9)$$

Proof. Again by sub-multiplicativity:

$$\|\mathbf{A}^{-1}\mathbf{F}^*(\hat{\mathbf{u}} - \hat{\mathbf{u}}^{\text{pred}})\| \leq \|\mathbf{A}^{-1}\| \cdot \|\mathbf{F}^*\| \cdot \|\hat{\mathbf{u}} - \hat{\mathbf{u}}^{\text{pred}}\|.$$

Since $\|\mathbf{F}^*\| = \|\mathbf{F}\|$ (operator norm equals largest singular value, which is the same for $\mathbf{F}$ and $\mathbf{F}^*$), we obtain (9). $\square$

3.4 Main result

Combining (6), Lemma 3.1, and Lemma 3.2 via the triangle inequality:

Theorem 3.3 (Bias–variance bound). *For any $\theta \in \Theta$, the reconstruction error satisfies*

$$\|\mathbf{v}(\theta) - \mathbf{v}^*(\theta)\| \leq \underbrace{\frac{\|(\alpha_t\mathbf{D}_t^*\mathbf{D}_t + \lambda\mathbf{I})\mathbf{v}(\theta)\|}{\sigma_{\min}(\mathbf{A})}}_{\text{regularisation bias}} + \underbrace{\frac{\|\mathbf{F}\|}{\sigma_{\min}(\mathbf{A})} \cdot \|\hat{\mathbf{u}}(\theta) - \hat{\mathbf{u}}^{\text{pred}}(\theta)\|_{\mathbb{C}^N}}_{\text{propagated surrogate error}}. \quad (10)$$

Taking expectations over θ and using Jensen’s inequality ($\mathbb{E}[\|\cdot\|] \leq \sqrt{\mathbb{E}[\|\cdot\|^2]}$ in the simpler L^1 form):

Corollary 3.4 (Expected error bound).

$$\mathcal{E}_{\text{tot}} \leq \frac{\mathbb{E}_{\theta}[\|(\alpha_t\mathbf{D}_t^*\mathbf{D}_t + \lambda\mathbf{I})\mathbf{v}(\theta)\|]}{\sigma_{\min}(\mathbf{A})} + \frac{\|\mathbf{F}\|}{\sigma_{\min}(\mathbf{A})} \cdot \mathbb{E}_{\theta}[\|\hat{\mathbf{u}}(\theta) - \hat{\mathbf{u}}^{\text{pred}}(\theta)\|_{\mathbb{C}^N}]. \quad (11)$$

4 Interpretation

4.1 Role of each term

Inequality (11) separates the total error into two independent contributions.

Regularisation bias. The numerator $\|(\alpha_t\mathbf{D}_t^*\mathbf{D}_t + \lambda\mathbf{I})\mathbf{v}\|$ measures how much the true solution deviates from the prior encoded by the regularisation:

- $\|\mathbf{D}_t\mathbf{v}\|^2 = \sum_n (v_{n+1} - v_n)^2$ is large when $\mathbf{v}$ varies rapidly in time — e.g. during the explosive early transient of the CH_4 dispersion.
- $\lambda\|\mathbf{v}\|$ penalises large amplitudes.

Increasing α_t or λ reduces $\sigma_{\min}(\mathbf{A})^{-1}$ (the denominator grows) but simultaneously increases the numerator. This is the standard *bias–variance trade-off*.

Propagated surrogate error. The factor $\|\mathbf{F}\|/\sigma_{\min}(\mathbf{A})$ is the *condition number of the inversion* with respect to the surrogate error. It quantifies how errors in the predicted Laplace coefficients $\hat{\mathbf{u}}^{\text{pred}}$ are amplified into errors on the reconstructed field $\mathbf{v}^*$.

Two observations:

1. $\|\mathbf{F}\|$ depends on the choice of frequencies $\{s_k\}$ — it equals the largest singular value of the Laplace matrix and is directly controlled by the contour geometry.
2. $\sigma_{\min}(\mathbf{A}) \geq \sigma_{\min}(\mathbf{F}^*\mathbf{F}) + \alpha_t \sigma_{\min}(\mathbf{D}_t^*\mathbf{D}_t) + \lambda$, so regularisation improves the condition number.

4.2 What remains to be bounded

Theorem 3.3 is fully rigorous. The only term that is not yet characterised is $\mathbb{E}_\theta[\|\hat{\mathbf{u}}(\theta) - \hat{\mathbf{u}}^{\text{pred}}(\theta)\|]$ — the mean surrogate error on the Laplace coefficients. Bounding this requires additional assumptions on the surrogate model (autoencoder + MLP), which are stated separately as working hypotheses and validated empirically.

Remark 4.1. *The bound (10) holds for any set of frequencies $\{s_k\}$ and any regularisation parameters (α_t, λ) , provided $\lambda > 0$. Optimising over these choices is the subject of future work.*

5 Comparison with the Bromwich–FFT Pipeline

We now apply the general framework to the pipeline used in the companion report [1], which combines a truncated Bromwich contour with explicit FFT inversion. This allows us to (i) compute the quantities $\|\mathbf{F}\|$ and $\sigma_{\min}(\mathbf{A})$ in closed form, and (ii) identify precisely what the truncated FFT does *not* capture within the bias–variance framework.

5.1 Bromwich contour and FFT structure

The Bromwich contour with damping $\gamma \geq 0$ uses N_t equally-spaced frequencies

$$s_k = \gamma + i\omega_k, \quad \omega_k = \frac{2\pi k}{N_t \Delta t}, \quad k = 0, \dots, N_t - 1.$$

The Laplace matrix $\mathbf{F} \in \mathbb{C}^{N_t \times N_t}$ factorises as

$$\mathbf{F} = \mathbf{D}_{\text{FFT}} \cdot \mathbf{W}, \tag{12}$$

where $\mathbf{D}_{\text{FFT}}$ is the $N_t \times N_t$ DFT matrix with $(D_{\text{FFT}})_{k,n} = e^{-i\omega_k t_n}$, and $\mathbf{W} = \text{diag}(\Delta t w_n e^{-\gamma t_n})$ encodes the quadrature weights and the exponential damping.

Since $\mathbf{D}_{\text{FFT}}^* \mathbf{D}_{\text{FFT}} = N_t \mathbf{I}$ (Parseval), we obtain

$$\mathbf{F}^* \mathbf{F} = \mathbf{W}^* \mathbf{D}_{\text{FFT}}^* \mathbf{D}_{\text{FFT}} \mathbf{W} = N_t \mathbf{W}^* \mathbf{W} = N_t \Delta t^2 \text{diag}(w_n^2 e^{-2\gamma t_n}). \tag{13}$$

This is *diagonal* — a key property of the Bromwich contour. It follows immediately that:

Proposition 5.1 (Spectral quantities on the full Bromwich contour). *On the full Bromwich contour with $N = N_t$ frequencies and damping $\gamma \in \mathbb{R}$, $\gamma > -\lambda_1$:*

$$\|\mathbf{F}\|^2 = \sigma_{\max}(\mathbf{F}^* \mathbf{F}) = N_t \Delta t^2 \max_n (w_n^2 e^{-2\gamma t_n}) = N_t \Delta t^2 \cdot \begin{cases} 1 & \gamma \geq 0, \\ e^{-2\gamma T} & \gamma < 0, \end{cases} \tag{14}$$

$$\sigma_{\min}(\mathbf{F}^* \mathbf{F}) = N_t \Delta t^2 \min_n (w_n^2 e^{-2\gamma t_n}) = N_t \Delta t^2 \cdot \begin{cases} e^{-2\gamma T} & \gamma \geq 0, \\ 1 & \gamma < 0. \end{cases} \tag{15}$$

In both cases the conditioning is

$$\kappa(\mathbf{F}^* \mathbf{F}) = e^{2|\gamma|T}. \tag{16}$$

Remark 5.2 (Role of γ on the full contour). Equation (16) shows that κ is minimised at $\gamma = 0$, where $\kappa = 1$ (perfect conditioning). Any deviation $\gamma \neq 0$ degrades conditioning exponentially in T .

The two signs of γ have asymmetric effects on the variance term $\|\mathbf{F}\|/\sigma_{\min}(\mathbf{A})$:

- $\gamma > 0$: $\|\mathbf{F}\| = \sqrt{N_t} \Delta t$ is unchanged (maximum at $t_0 = 0$), but $\sigma_{\min}$ decreases as $e^{-2\gamma T}$, so the ratio grows as $e^{\gamma T}$.
- $\gamma < 0$: $\|\mathbf{F}\|$ grows as $e^{|\gamma|T}$ (maximum at $t_{N_t-1} = T$), while $\sigma_{\min}$ is unchanged, so the ratio again grows as $e^{|\gamma|T}$.

Hence $\gamma = 0$ also minimises the variance amplification $\|\mathbf{F}\|/\sigma_{\min}$ on the full contour. However, as discussed in Section 5.2 below, the truncated contour introduces an additional spectral decay effect of γ that modifies this conclusion.

5.2 Truncation to $k_{\max}$ frequencies

In practice, the surrogate only predicts the first $k_{\max} \ll N_t$ Laplace modes. Let $\mathbf{F}_{k_{\max}} \in \mathbb{C}^{k_{\max} \times N_t}$ denote the truncated Laplace matrix (first $k_{\max}$ rows of $\mathbf{F}$). The Gram matrix $\mathbf{F}_{k_{\max}}^* \mathbf{F}_{k_{\max}}$ is no longer diagonal; it is a weighted Toeplitz matrix:

$$(\mathbf{F}_{k_{\max}}^* \mathbf{F}_{k_{\max}})_{n,n'} = N_t \Delta t^2 e^{-\gamma(t_n + t_{n'})} w_n w_{n'} D_{k_{\max}}(n - n'), \quad (17)$$

where $D_{k_{\max}}(m) = \sin(\pi k_{\max} m / N_t) / \sin(\pi m / N_t)$ is the Dirichlet kernel.

Proposition 5.3 (Spectral quantities after truncation). For the truncated Bromwich contour with $k_{\max}$ frequencies and damping γ :

$$\|\mathbf{F}_{k_{\max}}\|^2 \approx k_{\max} \Delta t^2 e^{-2\gamma t_{\min}} = k_{\max} \Delta t^2 \cdot \begin{cases} 1 & \gamma \geq 0, \\ e^{-2\gamma T} & \gamma < 0, \end{cases} \quad (18)$$

$$\sigma_{\min}(\mathbf{F}_{k_{\max}}^* \mathbf{F}_{k_{\max}}) \approx k_{\max} \Delta t^2 e^{-2\gamma T} \cdot \begin{cases} 1 & \gamma \geq 0, \\ e^{2\gamma T} & \gamma < 0, \end{cases} \quad (19)$$

so that the conditioning is $\kappa \approx e^{2|\gamma|T}$, as on the full contour.

Remark 5.4 (Trade-off induced by γ on the truncated contour). The truncation introduces an effect of γ that is absent on the full contour: it controls the spectral decay of the Laplace coefficients. The energy of the k -th Laplace mode decays as $\|\hat{u}_k\|^2 \sim \|U_0\|^2 e^{-2\gamma T} \omega_k^{-2}$, so:

- $\gamma > 0$ accelerates the decay of $\|\hat{u}_k\|$ with k , reducing the spectral truncation error (21) (fewer modes needed to represent the signal), at the cost of worsening conditioning $\kappa = e^{2\gamma T}$.
- $\gamma < 0$ slows spectral decay (more energy in high frequencies), requiring larger $k_{\max}$ to achieve the same truncation error, but improves conditioning.
- $\gamma = 0$ provides no spectral acceleration but achieves $\kappa = 1$, making the inversion perfectly conditioned.

The optimal γ therefore depends on the relative magnitude of the spectral truncation error and the propagated surrogate error, and is problem-dependent. This is consistent with the empirical finding in the companion report that $\gamma = 0$ is near-optimal for the CH_4 dataset, where $k_{\max} = 20$ already captures the dominant spectral energy.

Remark 5.5 (Effect of $k_{\max}$ at fixed γ). Both $\|\mathbf{F}_{k_{\max}}\|$ and $\sigma_{\min}$ scale as $k_{\max}$ at fixed γ , so their ratio

$$\frac{\|\mathbf{F}_{k_{\max}}\|}{\sigma_{\min}(\mathbf{A})} \approx \frac{1}{\sqrt{k_{\max}} \Delta t e^{-\gamma T}}$$

decreases with $k_{\max}$ — the variance amplification improves. However, the surrogate error $\mathbb{E}_{\theta}[\|\hat{u} - \hat{u}^{\text{pred}}\|]$ grows with $k_{\max}$, creating a trade-off that depends on the surrogate model capacity.

5.3 The FFT inversion is not the regularised inverse

A critical observation is that the pipeline in the companion report does *not* solve the regularised least-squares problem (3). Instead, it applies the explicit FFT inversion formula:

$$v_n^* = \frac{e^{\gamma t_n}}{\Delta t} \sum_{k=0}^{k_{\max}-1} \hat{u}_k^{\text{pred}} e^{i\omega_k t_n} + \frac{e^{\gamma t_n}}{\Delta t} \sum_{k=k_{\max}}^{N_t-1} \bar{U}_k e^{i\omega_k t_n}, \quad (20)$$

where $\bar{U}_k$ is the per-pixel training mean for $k > k_{\max}$.

This operation corresponds to $\alpha_t = \lambda = 0$ in (3) *together with an explicit spectral truncation*. These are two distinct sources of error that do not appear in the bias–variance decomposition of Theorem 3.3:

1. **Spectral truncation error.** Even with a perfect surrogate ($\hat{u}^{\text{pred}} = \hat{u}$), the FFT reconstruction omits the high-frequency content:

$$\mathbf{v} - \mathbf{v}_{\text{FFT}}^* = \frac{e^{\gamma t_n}}{\Delta t} \sum_{k=k_{\max}}^{N_t-1} (\hat{u}_k - \bar{U}_k) e^{i\omega_k t_n}. \quad (21)$$

This is the source of the Gibbs oscillations reported in the companion paper, which required a post-hoc UNet corrector to suppress.

2. **Absence of regularisation bias.** Setting $\alpha_t = \lambda = 0$ gives zero regularisation bias in (10). However, this comes at the cost of the spectral truncation error (21), which is not controlled by the framework.

5.4 Summary comparison

Table 1 summarises the two pipelines within the bias–variance framework.

	FFT truncation (companion report)	Regularised inverse (this work)
Inversion formula	Explicit: $e^{\gamma t}/\Delta t \cdot \text{FFT}^{-1}$	$\mathbf{A}^{-1} \mathbf{F}^* \hat{\mathbf{u}}^{\text{pred}}$
α_t, λ	0	> 0
Regularisation bias	0 (within framework)	$\alpha_t \ \mathbf{D}_t \mathbf{v}\ / \sigma_{\min}(\mathbf{A})$
Spectral truncation error	Yes — source of Gibbs oscillations	No
Requires UNet corrector	Yes (481k parameters)	No
$\ \mathbf{F}\ $	$\sqrt{k_{\max}} \Delta t$	depends on contour Γ
$\sigma_{\min}(\mathbf{A})$	$k_{\max} \Delta t^2$	$\sigma_{\min}(\mathbf{F}^* \mathbf{F}) + \alpha_t \pi^2 / N_t^2 + \lambda$

Table 1: Comparison of the FFT truncation pipeline and the regularised inverse within the bias–variance framework of Theorem 3.3.

The key insight is that the spectral truncation error (21) falls *outside* the bias–variance decomposition of Theorem 3.3: it is neither a regularisation bias nor a propagated surrogate error. It is an intrinsic limitation of the FFT inversion that the regularised inverse eliminates by construction, at the cost of introducing a controlled regularisation bias.

References

- [1] K. Attarian, *Learning Latent Representations for Neural Surrogate Models of Partial Differential Equations*, École Polytechnique / Institut Polytechnique de Paris, April 2026.